

Advance python (1).ipynb

if
else
if else
nested if
if elif else

```
if True: # indentation is always 4 spaces
    print('Data Science')
```

```
if True:
    print('ds')
```

```
if False:
    print('Data Science')
print('bye for now')
```

```
if True: # indentation is always 4 spaces
    print('Data Science')
print('bye for now')
```

```
if False: # indentation is always 4 spaces
    print('Data Science')
else:
    print('bye for now')
```

Lets do one program as if divide by 2 then reminder is 0 then it is even number
if reminder is not 0 then it is odd number

```
#to print only even number
x = 4
r = x % 2
```

```
if r == 0:
    print('Even number')
```

```
#to print only even number
x = 5
r = x % 2
```

```
if r == 0:
```

```
print('Even number')
```

```
x = 5
```

```
r = x % 2
```

```
if r == 0:
```

```
    print('Even number')
```

```
print('odd number')
```

```
x = 6
```

```
r = x % 2
```

```
if r == 0:
```

```
    print('Even number')
```

```
print('odd number')
```

```
x = 10
```

```
r = x % 2
```

```
if r == 0:
```

```
    print('Even number')
```

```
if r == 1:
```

```
    print('odd number')
```

```
x = 11
```

```
r = x % 2
```

```
if r == 0:
```

```
    print('Even number')
```

```
if r == 1:
```

```
    print('odd number')
```

```
x = 23
```

```
r = x % 2
```

```
if r == 0:
```

```
    print('Even number')
```

```
if r == 1:
```

```
    print('odd number')
```

```
x = 17
```

```
r = x % 2
```

```
if r == 0:
    print('Even number')

if r != 0:
    print('odd number')
```

if we observe the code its too many line cuz many of the coder always they wanted to reduce the code lenght which is very good practise. instead of 2 if we can use if-- else

```
x = 5
r = x % 2

if r == 0:
    print('Even number')
else:
    print('Odd Number')
```

```
x = 4
r = x % 2

if r == 0:
    print('Even number')
else:
    print('Odd Number')
```

NESTED IF (if we have 2 condition so we need to implment with nested if)

```
x = 3
r = x % 2

if r == 0:
    print('Even number')
    if x>5:
        print('greater number')

else:
    print('Odd Number')
```

```
x = 4
r = x % 2

if r == 0:
    print('Even number')
```

```
    if x>5:
        print('greater number')

else:
    print('Odd Number')
```

We do have concept of (IF - ELIF- ELSE)

e.g i want to print (1--> one , 2 --> two, 3--> three, 4--> four, 5- five)

```
# when you use if it will check all condition but if we mention elif
then it wont check all condition
# when we use if condition it will check all every block of code
better debug in pycharm
# you can debug with value 1 & d for both if & elif
```

```
x = 2
```

```
if x == 1:
    print('one')
if x == 2:
    print('Two')
if x == 3:
    print('Three')
if x == 4:
    print('four')
```

```
# elif it wont check till the block once you find the output it wont
go to next line
# you can try with multiple parameter 1, 2 & 3 value in x
```

```
x = 4
```

```
if(x == 1):
    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')
```

```
x = 7
```

```
if(x == 1):
```

```

    print('one')
elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')

x = 7

if(x == 1):
    print('one')

elif(x == 2):
    print('Two')
elif(x == 3):
    print('Three')
elif(x == 4):
    print('four')

else:
    print('wrong output')

#short hand if
a = 30
b = 20
if a > b: print("a is greater than b")

```

LOOPS -- in programing world some time we keep on repeating , may be you want to repeat 5 statement so one way is copy & paste multiple times or other way is

if you want to print the datascience 1000 times then what you will you cant copy for 1000 times , if you want to print 1000 times then you cant do manually . that is the reason why we need to apply loop ->

2 type of loops -- While loop & For loop

```

print('data science')
print('data science')
print('data science')
print('data science')
print('data science')

```

```
i = 1          # initializing
```

```
while i<=5:    # condition
    print('data science')
    i = i + 1  # increment
```

```
i = 5          # initializing
```

```
while i>=1:    # condition
    print('data science')
    i = i - 1  # decrement
```

```
i = 1          # initializing
while i<=5:    # condition
    print('data science : ', i)
    i = i + 1  # increment
```

```
i = 5          # initializing
while i>=1:    # condition
    print('data science : ',i)
    i = i - 1  # decrement
```

can we use multiple while loop || nested while loop

to understand nested while indepth understand you can use pycharm debug with f8 option

```
i = 1
```

```
while i<=5:
    print('data science') # when we mention end then new line will not
create
```

```
    j = 1
    while j<=4:
        print('technology')
        j = j + 1
```

```
    i = i + 1
    print()
```

the output which we got is very lengthy but how to make them one line lets refer to below code

```
i = 1
while i<=5:
```

```
print(' datascience', end = "") # when we mention end then new
line will not create
```

```
j = 1
while j<=4:
    print(' technology', end="")
    j = j + 1
```

```
i = i + 1
print()
```

```
# lets use while loop usig some numbers
i = 1
```

```
while i <= 2 :
    j = 0
    while j <= 2 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

```
# lets use while loop usig some numbers
i = 1
while i <= 4 :
    j = 0
    while j <= 3 :
        print(i*j, end=" ")
        j += 1
    print()
    i += 1
```

FOR LOOP -

normally while loop it work with iteration or certaion some condition but for loop it will work with sequence (list, string,int)

```
name = 'nit'
```

```
for i in name:
    print(i)
```

```
name1 = [1,3.5,'hallo'] #i want print the value individualy
```

```
for i in name1:
```

```
print(i)
```

```
for i in [2, 3, 7.8, 'hi']:  
    print(i)
```

```
range(5)
```

```
for i in range(5):  
    print(i)
```

```
for i in range(2,5):  
    print(i)
```

```
for i in range(1,10,3):  
    print(i)
```

```
# print the value which is divisible by 5  
for i in range(1,21):  
    print(i)
```

```
# print the value which is divisible by 5
```

```
for i in range(1,51):
```

```
    if i%5==0 :  
        print(i)
```

```
# print the value which is divisible by 5 i dont want that value  
for i in range(1,51):
```

```
    if i%5!=0 :  
        print(i)
```

```
# LETS DISCUSS ABOUT 3 KEYWORDS -- BREAK || CONTINUE || PASS
```

```
BREAK STATEMNT - if you apply break statment in a loop then it will end the loop
```

```
# Pass = skips block of code( function, class etc)
```

```
# Continue= skips 1 step/iteration during loop
```

```
# Break= jumps out of the function/loop
```

```
# write the code user ask chocklet from vendor machne write the basic  
code
```



```
x = int(input('How many choclets you want:?'))
```

```
i = 1
while i<=x:
    print('choclet')
    i += 1
```

- If the user says i need 10 choclet but vending machine dont have 10 chocolate & machine has only 5 chocolate so what you do on those scenario

- We have 3 choice now (eiter stop the transaction by you or you can give only 5 chocolate) & may be

vendor machine display the result as we are out of the stock

- Now lets try in the code

```
ava = 5 # the machine has only 5 choclet
```

```
x = int(input('How many choclets you want:?'))
```

```
i = 1
while i<=x:
    print('choclet')
    i += 1
```

if you check the user wants 10 choclets but availabe choclet is 5 but we got output as 10 choclet

in this code we just declare but we dint apply any condition to it

```
available_choclet = 5 # the machine has only 10 candis
```

```
x = int(input('How many choclets user want:?'))
```

```
i = 1
while i<=x:
```

```
    if i>available_choclet: # we stop the execution but which code
        execution not entire code , i want to come out of while loop
```

```
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1
```

```
print('bye for now')
```

```
available_choclet = 5 # the machine has only 10 candis
```

```
x = int(input('How many choclets you want:?'))
```

```
i = 1
while i<=x:

    if i>available_choclet: # we stop the execution but which code
execution not entire code , i want to come out of while loop
        print('out of stock')
        break # break is statement | means jump out of the loop
    print('choclet')
    i += 1
```

```
print('bye for now')
```

```
for i in range(1,11):
    print(i)
```

- i dont want 11 number i want only 5 number for the range of 1 to 10

```
for i in range(1,11):
    if i == 6:
        break
    print(i)
```

in continue loop wont be terminate & exclue the assign number it give you entire output

```
for i in range(1,11):
    if i == 3:
        continue
    print(i)
```

```
for i in range(1,11):
    if i == 6:
        continue
    print('hello : ',i)
```

#PASS Statement - pass the code & it wont go (code give you the error)

```
for i in range(1,11):
```

```
for i in range(1,11):
    pass
```

you need to print the number from 1 to 50 but dont print the number which is divisible by 3 or 5

```
for i in range(1,51):
```

```
    if i%3 == 0:
```

```
        print(i)
```

```
print('end')
```

```
for i in range(1,51):
```

```
    if i%3 == 0:
```

```
        continue
```

```
    print(i)
```

```
print('end')
```

```
for i in range(1,51):
```

```
    if i%3 == 0 or i%5 == 0:
```

```
        continue
```

```
    print(i)
```

```
#print('end')
```

```
# it will skip all the value which is divisible by 3 or 5
```

```
for i in range(1,50):
```

```
    if i%3 == 0 or i%5 == 0:
```

```
        continue
```

```
    print(i)
```

```
print('end')
```

```
# when you apply and you wont get the value which is divisible by both  
3 & 5 (15)
```

```
# i dont want to print the values which are even numbers that means  
print only odd numbers
```

```
for i in range(1,51):
```

```
    if (i%2 == 0):
```

```
        #print('even')
```

```
        continue
```

```
    else:
```

```
        print(i)
```

```
print('bye')
```

```
# PRINTING PATTERN IN PYTHON
```

```
print('# # # #')  
print('# # # #')  
print('# # # #')  
print('# # # #')
```

```
for i in range(1,5):  
    i=i+1  
    print('# # # # ')
```

```
for i in range(1,5):  
    if i<=5:  
        print('# # # #')
```

```
for j in range(4):  
    print('#')
```

```
for j in range(4):  
    print('# # # #')
```

```
for j in range(4):  
    print('#', end = " ")
```

```
for j in range(4):  
    print('#', end=" ")
```

```
for j in range(4):  
    print('#', end=" ")
```

```
for j in range(4):  
    print('#', end=" ")
```

```
print()
```

```
for j in range(4):  
    print('#', end=" ")
```

```
for j in range(4):
```

```
    print('#', end=" ")

print()

for j in range(4):
    print('#', end=" ")

print()

for j in range(4):
    print('#', end=" ")

print()

for j in range(4):
    print('#', end=" ")

for i in range(4):
    for j in range(4):
        print('#', end=" ")
    print()
    # pease use debug mode in pycharm

for i in range(4):
    for j in range(i+1):
        print('#', end = " ")
    print()

for i in range(1,5):
    print("# " * i)

for i in range(1,5):
    for j in range(4):
        if i > j:
            print("#", end=" ")
    print()

list(range(5))

for i in range(4):
    for j in range(i):
        print('#', end=" ")
    print()
```

```
for i in range(4):
    for j in range(i+1):
        print('#', end=" ")
    print()
```

```
for i in range(4):
    for j in range(4-i):
        print('#', end=" ")
    print()
```

```
for i in range(1,5):
    print("# "*(5-i))
```

- For|Else in python
- In other language for else not supportable but in python it is supportable

eg- lets print the number from 1- 20 & we dont want print number which is divisible by 5

```
nums = [12,15,18,21,26, 30, 40]
```

```
for num in nums:
    if num % 5 == 0:
        print(num)
```

```
nums = [12,14,18,21,25,30,35]
```

```
for num in nums:
    if num % 5 == 0:
        print(num)
```

```
nums = [12,14,18,21,25,20]
```

```
for num in nums:
    if num % 5 == 0:
        print(num)
```

```
nums = [12,14,18,21,20,25]
```

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

```
nums = [12,14,18,21,20,25]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break
```

```
nums = [10,14,18,21,5,10]
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break #it will print only 1 number then it break
```

```
nums = [7,14,18,21,23,27] #hear there is no number which is divisible  
by 5 we got output as blank
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break
```

```
nums = [7,14,18,21,23,27,29] #hear there is no number which is  
divisible by 5 we got output as blank
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break  
    else:  
        print('Number Not Found') #every iteration it cheking  
condition
```

```
nums = [7,14] #hear there is no number which is divisible by 5 we got  
output as blank
```

```
for num in nums:  
    if num % 5 == 0:  
        print(num)  
        break  
    else:  
        print('Number Not Found') #every iteration it cheking  
condition
```

nums = [7,14,18,21,23,27] #hear there is no number which is divisible by 5 we got output as blank

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Number Not Found') # hear else we dont write in if
block but we can write in for block only
```

nums = [10,14,18,21,20,27] #hear there is no number which is divisible by 5 we got output as blank

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Not Found')
```

nums = [10,14,18,21,20,27,30] #hear there is no number which is divisible by 5 we got output as blank

```
for num in nums:
    if num % 5 == 0:
        print(num)
        #break
else:
    print('Not Found')
```

nums = [10,14,18,21,20,27] #hear there is no number which is divisible by 5 we got output as blank

```
for num in nums:
    if num % 5 == 0:
        print(num)
        break
else:
    print('Not Found')
```

- prime number - how to check given number is prime number or not


```
num = 14
```

```
for i in range(2,num):  
    if num % i == 0:  
        print('Not prime Number')  
        break  
else:  
    print('Prime Number')
```

```
num = 13
```

```
for i in range(2,num):  
    if num % i == 0:  
        print('Not prime Number')  
        break  
else:  
    print('Prime Number')
```

Array in python

```
from array import *  
arr = array('i',[])
```

```
n = int(input('Enter the length of the array'))
```

```
for i in range(5):  
    x = int(input('Enter the next value'))  
    arr.append(x)  
print(arr)
```

```
from array import *  
arr = array('i',[])
```

```
n = int(input('Enter the length of the array'))
```

```
for i in range(5):  
    x = int(input('Enter the next value'))  
    arr.append(x)  
print(arr)
```

```
from array import *  
arr = array('i',[])
```

```
n = input('Enter the length of the array')
```

```
for i in range(5):  
    x = input('Enter the next value')  
    arr.append(x)  
print(arr)
```

Way of creating array using numpy

```
from numpy import *  
arr = array([1,2,3,4,5])  
print(arr)  
type(arr)
```

```
print(arr.dtype)
```

```
arr2 = array([1,2,3,4,5.9],float)  
arr2
```

```
arr3 = array([1,2,3,4,5.6],int)  
arr3
```

```
import numpy as np
```

```
arr4 = np.linspace(0, 16, 10) # break the code between 10 spaces  
between 0 to 16 but why decimal becuse we break into parts thats why  
it is floats  
arr4
```

```
arr5 = np.arange(0,10,2) # arange - as range  
arr5
```

```
arr6 = np.zeros(5)  
arr6
```

```
arr7 = np.ones(5)  
arr7
```

```
import tkinter as tk  
# Function to be called when the button is clicked
```

```
def on_button_click():
    label.config(text="Button clicked!")

# Create the main application window
root = tk.Tk()
root.title("Simple Tkinter App")

# Create a label widget
label = tk.Label(root, text="Hello, Tkinter!")
label.pack(pady=20)

# Create a button widget
button = tk.Button(root, text="Click Me", command=on_button_click)
button.pack(pady=20)

# Run the application
root.mainloop()
```